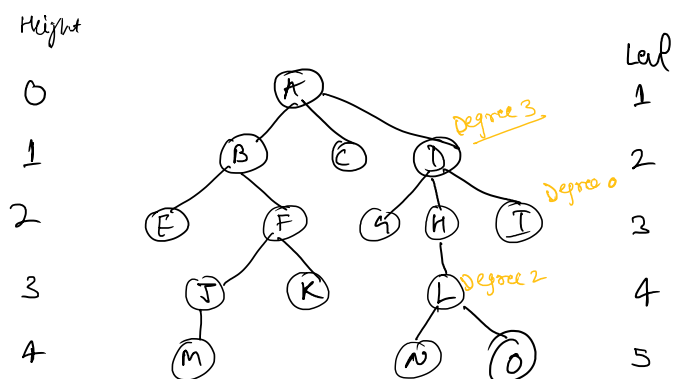


# Tree

15 June 2025 13:33



forest :- Collection of trees

Degree :- no of child nodes

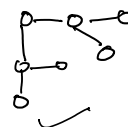
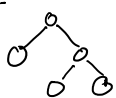
Internal nodes :- which has atleast 1 node

External nodes :- Also known as leaf node

## Binary tree

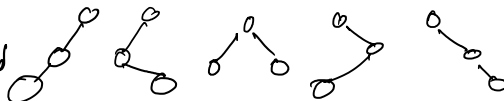
It has max 2 nodes as child

examples



## Number of binary tree

Let's say  $n=3$  So 5 trees can be generated



$$T(3) = 5$$

$$T(4) = 14$$

$$T(n) = \frac{2^n C_n}{n+1} \quad \text{Catalan Number}$$

$$T(5) = \frac{2^5 C_5}{5+1} = \frac{10 C_5}{6} \Rightarrow \frac{10 \times 9 \times 8 \times 7 \times 6}{5 \times 4 \times 3 \times 2 \times 1} \Rightarrow 42$$

no. of trees with max height  $\Rightarrow 2^{n-1}$

$$n=3 = \text{height}=2 \Rightarrow 4 \text{ trees} \Rightarrow 2^2$$

$$n=4 \Rightarrow 3 \Rightarrow 8 \text{ trees} \Rightarrow 2^3$$

$$n=5 \Rightarrow 4 \Rightarrow 16 \text{ trees} \Rightarrow 2^4$$

### Another way of Catalan formula

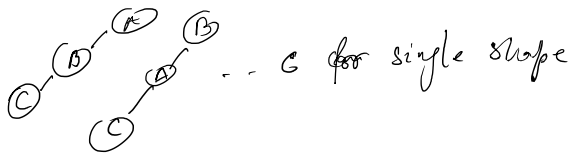
$$T(6) = T(0) \times T(5) + T(1) \times T(4) + T(2) \times T(3) + T(3) \times T(2) + T(4) \times T(1) + T(5) \times T(0)$$

$$= \sum_{i=0}^n T(i-1) \times T(n-i)$$

There are two types of nodes labelled and unlabelled

Above we saw unlabelled nodes examples as they haven't mentioned anything in nodes

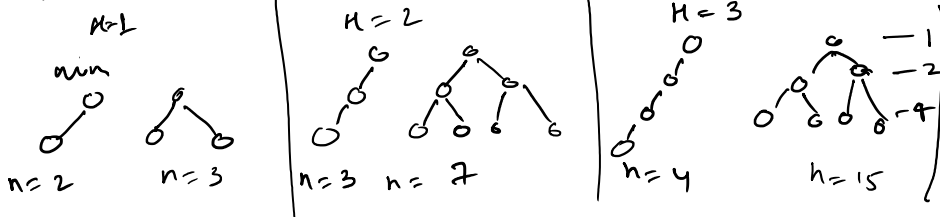
Labelled nodes  $n=3$  (A) (B) (C)



So total  $\Rightarrow T(n) = \frac{2^n C_n}{n+1} \times n!$

### Height vs Nodes

min Nodes and Max Nodes



min nodes = Height + 1

max nodes = GP series =  $a + ar + ar^2 + \dots + ar^k$

$$= \frac{a(r^{k+1} - 1)}{r - 1}$$

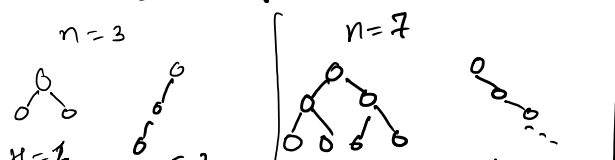
$$= \frac{2(2^{h+1} - 1)}{2 - 1} = \underline{2^{h+1} - 1}$$

$$\text{ex } H=3 = 2^{3+1} - 1$$

$$= 16 - 1$$

$$= 15$$

nodes given find min & max height



$$\log_2(n+1) - 1 \leq H \leq n-1$$

Height of Binary tree can vary from  $O(\log n)$  to  $O(n)$

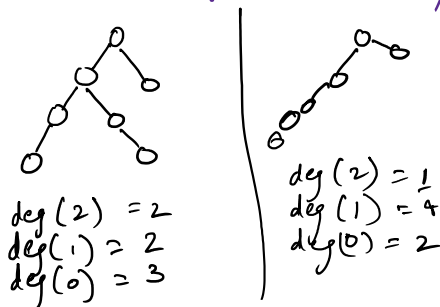
... |  $H=2$  |  $H=6$  |

max Height = Height - 1

Min Height =  $n = 2^{H+1} - 1 \Rightarrow n+1 = 2^{H+1} \Rightarrow \log^{(n+1)} = H+1$   
 $\Rightarrow \log^{(n+1)} - 1 = H$

=  $\log^{(n+1)} - 1$

### Relationship between Leaf node and normal node



$deg(0) = deg(2) + 1$  for Binary tree

### Strict Binary Tree

Tree having zero or 2 children



if  $h$  is given then

min nodes =  $2 \times h + 1$

max nodes =  $2^{H+1} - 1$

or degree  $\times$  height + 1

or  $\frac{m^{H+1} - 1}{m - 1}$

$\therefore$  Binary = 2 degree  
 Binary = 3 degree  
 4-ary = 4 degree

if  $n$  is given then

min height =  $\log_2^{(n+1)} - 1$

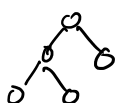
or  $\log_m^{[n(m-1)+1]} - 1$

where  $m = \text{degree}$

max height =  $(n-1)/2$

or  $\frac{(n-1)}{m}$

Internal nodes V/s External nodes



$e = 3$   
 $i = 2$

$e = i + 1$

or  $(m-1)i + 1$